

Experience

Apple ▪ Senior Software Engineer (Contract)

System Intelligence Team, Machine Learning and AI Group March 2024 – Sep 2025

- Productionized research prototype Talaria supporting **Apple foundation model latency optimization** using Flask, Vue, Svelte, and K8s, improving on-device inference efficiency of **50+ machine learning vision models**
- **Led design and development** of internal Next.js storefronts for configuring, integrating, and sharing LLMs
- Expanded interactive graph visualization tool Mycelium (open-source), a TypeScript and Svelte library used for visualizing deep neural networks and Transformer architectures; implemented WebGL viewer achieving **5x higher** framerates when viewing LLMs, and enhanced feature set for models running on Apple Neural Engine
- Engineered Python library enabling researchers to compile models to hardware with **high reliability** ($\geq 99.9\%$)

Memorial Sloan Kettering Cancer Center ▪ Senior Software Engineer

Pathology Results Team, Clinical Bioinformatics Group May 2018 – Present

- **Tech lead** (8 developers) for MPath, a clinical pathology software for displaying genomic testing results and matching patients to targeted therapies; established roadmaps and **collaborations across 5 teams**
- Architected and implemented performant web apps for managing and visualizing high-throughput clinical data supporting **120k+ patients**, using JavaScript, React, Python, Flask, MySQL, and MongoDB
- **Redesigned and rewrote core UIs**, improving query times, performance metrics, code splitting, end-to-end testing, state machines, and application monitoring, achieving **24% faster clinical report turnaround times**
- Spearheaded development lifecycle by creating build automations, design system, and CI/CD pipelines for **40+ developers** to achieve code optimization, visual unity, and standardized tooling across 10+ portals
- Developed large ETL systems and APIs delivering **2x faster** processing times using Python, Airflow, and K8s

Skills

Languages ▪ JavaScript, TypeScript, Python, Ruby, SQL, HTML, CSS, Sass

Libraries ▪ React, Flask, FastAPI, Next.js, Vue, Tailwind, Node.js, Playwright, Cypress, pandas

Tools ▪ Git, Docker, GitLab CI/CD, GitHub Actions, Apache Airflow, nginx, AWS S3, ELK, K8s

Design ▪ Figma, Adobe Photoshop, Lightroom, Illustrator

Education

Macaulay Honors College New York, NY

B.A. Chemistry and Bioinformatics, Minor in Mathematics, GPA 3.95 Aug 2013 – May 2017

- Data Science Fellowship 2016
- Full Merit Scholarship 2013 – 2017

Publications

Khwaja E, Song YS, **Agarunov A**, Huang B. CELL-E 2: Translating Proteins to Pictures and Back with a Bidirectional Text-to-Image Transformer. **NeurIPS** (2023).

Brannon AR, [et al, including **Agarunov A**]. Enhanced specificity of clinical high-sensitivity tumor mutation profiling in cfDNA via paired normal sequencing using MSK-ACCESS. **Nature Communications**. (2021).

Projects

Turbokit ▪ Framework for building production-ready, scalable, monorepo microfrontend React apps

ronivonu ▪ Generative art and creative coding projects created in JavaScript, Processing, and Python